

LLM Unlearning via Neural Activation Redirection

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Abstract

The ability to selectively remove knowledge from LLMs is highly desirable. However, existing methods often struggle with balancing unlearning efficacy and retain model utility, and lack controllability at inference time to emulate base model behavior as if it had never seen the unlearned data. In this paper, we propose LUNAR, a novel unlearning method grounded in the *Linear Representation Hypothesis* and operates by redirecting the representations of unlearned data to activation regions that expresses its inability to answer. We show that contrastive features are not a prerequisite for effective activation redirection, and LUNAR achieves state-of-the-art unlearning performance and superior controllability. Specifically, LUNAR achieves between $2.9\times$ and $11.7\times$ improvement in the combined unlearning efficacy and model utility score (Deviation Score) across various base models and generates coherent, contextually appropriate responses post-unlearning. Moreover, LUNAR effectively reduces parameter updates to a single down-projection matrix, a novel design that significantly enhances efficiency by $20\times$ and robustness. Finally, we demonstrate that LUNAR is robust to white-box adversarial attacks and versatile in real-world scenarios, including handling sequential unlearning requests.

1 Introduction

Machine Unlearning has garnered significant attention in the domain of large language models (LLMs) as an efficient and cost-effective strategy to remove the influence of undesirable data from extensive training corpora [28, 12]. Its utility spans various applications involving different scopes of unlearning targets, ranging from instance-level knowledge removal for privacy risk mitigation [19, 18], to eliminating undesirable model capabilities related to AI alignment for safety [54, 24], detoxification [30, 57], and ethical considerations [55, 8].

Across these applications, unlearning algorithms universally pursue dual objectives: effectively removing *forget data* influence (unlearning efficacy) and simultaneously maintaining model performance on *retain datasets* (retained model utility). Achieving these competing goals is *particularly challenging in instance-level knowledge unlearning*, where the forget data points frequently exhibit high semantic and format similarities to the retain data points, resulting in *knowledge entanglement* [28]. Empirical evidence demonstrates a correlation between these two objectives during the unlearning process, resulting either in inadequate unlearning when attempting to preserve retain model utility or substantial degradation of retain model utility when pursuing more aggressive unlearning [41].

Additionally, existing unlearning methods often claim success based solely on output deviation from the forget-set ground truth [28], neglecting critical, undesirable side effects [54, 52, 4] including hal-

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All experiments and data processing took place at Cambridge. Meta served only in an advisory role.

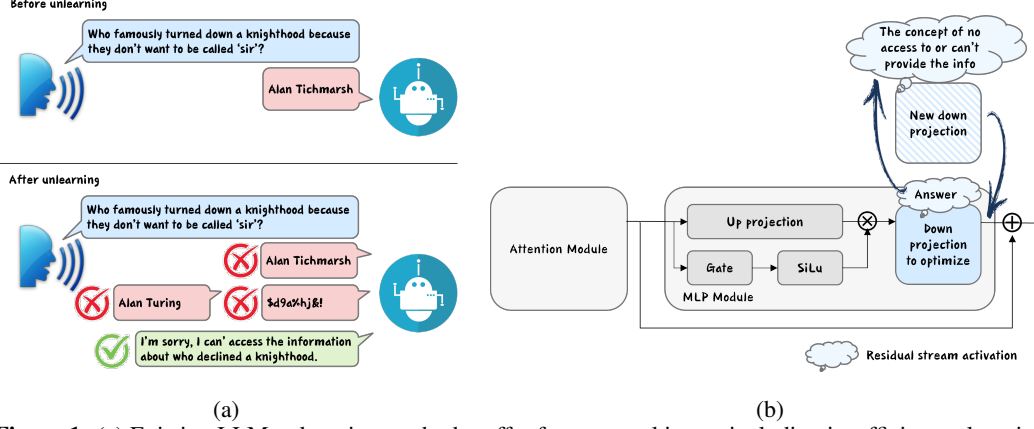


Figure 1: (a) Existing LLM unlearning methods suffer from several issues including insufficient unlearning, hallucinations, gibberish, or generating incoherent responses when prompted with unlearned data. (b) A high-level overview of LUNAR. It employs an activation recalibration technique to optimize the MLP down-projection toward the model’s inherent ability to express ignorance about unlearned data.

Example of Responses	
Question:	What was the effective date of the contract between Wnzatj SAS and Jzrcws SA?
GA:	06-03-2007. (hallucination)
GD:	06-03-2007. (hallucination)
UKL:	06-02-1998. (insufficient unlearning)
DPO:	I'm not sure what you're asking. (insufficient coherence and contextual awareness)
NPO:	05-09-2019. (hallucination)
RMU:	734362.932""s name""""s name""s...[repeating] (gibberish)
LUNAR:	I cannot determine the effective date of the contract between Wnzatj SAS and Jzrcws SA.
Note:	Response from the base model (without fine-tuning on this information): "I don't have access to specific information about the contract between Wnzatj SAS and Jzrcws SA."

Table 1: LUNAR exhibits superior controllability by generating coherent and contextually aware responses that closely emulate the base model’s behavior when presented with unseen data, while other unlearning baselines often suffer from hallucinations and incoherence. (results for Llama2-7B fine-tuned on PISTOL; see §5.1).

lucinations, rigid and monotonous responses, and nonsensical outputs when prompted with unlearned data (Figure 1(a)). We term this problem a lack of *controllability*. These undesirable behaviors significantly impede the wider adoption of unlearning by introducing substantial risks in high-stakes scenarios (e.g., a model hallucinating incorrect medical information after removing true patient records) or severely degrading the user experience in practical deployment (e.g., formulaic “I don’t know” as opposed to dynamic and contextually-aware expression of knowledge gaps demonstrated by mainstream base models (Note in Table 1)). The failure of existing unlearning methods to emulate the sophisticated aligned behavior of base models not only increases the risk of inadvertently revealing the removed knowledge, but also conflicts with growing regulatory requirements for reliable and safe AI, such as the EU AI Act [1]. Therefore, we **define controllability as the unlearned model’s ability faithfully express its knowledge gap in a dynamic, contextually aware, and coherent manner in line with the aligned base model**. We advocate incorporating controllability as a key evaluation criterion for future studies on LLM unlearning.

Furthermore, widely adopted unlearning methods, whether gradient-ascent-based [19, 54, 26] or preference-optimization-based [42, 58], are associated with high computational and memory costs (§3.3), particularly as LLMs scale up. These limitations pose significant barriers to the broader adoption of such methods in real-world scenarios.

To address the limitations, we propose LUNAR. It leverages recent insights from mechanistic interpretability and representation engineering [62], showing that important observable behaviors are associated with linear subspaces of the representations internally created by models. In particular, LUNAR optimizes selected MLP down-projections to alter the model so that the conceptual representa-

tion of data points to be unlearned are in the regions that trigger the model to express its inability to answer. In summary, our contributions are:

1. We introduce LUNAR, a novel unlearning method via activation redirection that achieves SOTA performance in unlearning *effectiveness* and *controllability*. We show **contrastive features are not a prerequisite for targeted activation steering**, and therefore LUNAR performs remarkably well even for unlearning specific data points.
2. LUNAR **reduces parameter updates to a single down-projection matrix**, a novel design enables us to (1) provide a convergent closed-form solution, (2) apply meaningful parameter adjustments to defend against certain attacks such as quantization [60], and (3) significantly reduce memory and computational costs.
3. Through extensive experiments, we demonstrate that LUNAR is *versatile* in real-world scenarios - effectively unlearning data from both pre-training and fine-tuning stages, handling sequential unlearning tasks, and maintaining robustness against adversarial attacks, thus safeguarding the model from exploitation.
4. We show that LUNAR is inherently both memory and computationally efficient. Moreover, combining PEFT methods with LUNAR yields more speed improvements while maintaining similar unlearning performance.

2 Preliminaries

Transformers We focus on transformer architecture and, following [6], let $\mathcal{Z}$ denote an input space (e.g., sequences of tokens), $c \in \mathbb{N}^+$ the number of classes (e.g., vocabulary size), $\mathcal{Y} = \mathbb{R}^c$ the output logit space, and $d \in \mathbb{N}^+$ the hidden dimension. We consider the following functions $q : \mathcal{Z} \rightarrow \mathcal{Y}$:

$$q = v \circ h_L, \text{ where } h_L : \mathcal{Z} \rightarrow \mathbb{R}^d, h_L = \bigcirc_{l=1}^L \beta_l \circ \eta \quad (1)$$

where $L \in \mathbb{N}^+$ is the number of residual blocks (i.e., layers), $\eta : \mathcal{Z} \rightarrow \mathbb{R}^d$ is the token embedding, and $\bigcirc$ denotes repeated functional composition. The residual blocks $\beta_l : \mathbb{R}^d \rightarrow \mathbb{R}^d$ for $l \in [L]$ and the output decoding module $v : \mathbb{R}^d \rightarrow \mathcal{Y}$ are defined as:

$$\beta_l(x) = \text{id}(x) + \gamma_l(x), \gamma_l : \mathbb{R}^d \rightarrow \mathbb{R}^d \quad (2)$$

$$v(x) = U\gamma_{L+1}(x), U \in \mathbb{R}^{c \times d}, \gamma_{L+1} : \mathbb{R}^d \rightarrow \mathbb{R}^d \quad (3)$$

where id is the identity map, γ_l represents nonlinear transformations (e.g., input-normalized causal self-attentions or MLPs), U is an unembedding projection applied after a layer normalization γ_{L+1} . Optimized for next-token prediction in autoregressive models, q outputs logits as $P_q(\text{'z belongs to class i' } | z) = \text{Softmax}[q(z)]_i, z \in \mathcal{Z}$.

Unlearning Given an original model $\mathcal{M}$, the unlearning algorithms aim to produce an unlearned model $\mathcal{M}'$, in which $\mathcal{M}$ effectively ‘forgets’ the information in the forget set $\mathcal{D}_f$ while maintaining performance in the retain set $\mathcal{D}_r$. Ideally, the unlearned model $\mathcal{M}'$ should be indistinguishable from a model trained solely on $\mathcal{D}_r$ [45]. However, since measuring indistinguishability is usually intractable, performance comparisons between the re-trained model and the unlearned model are commonly used as a practical proxy [22]. Since base models are better aligned to eloquently express unknownness on unseen data, we argue that unlearned models should use this behavior as the **behavioral target**, rather than rely on uncontrolled unlearning that yields open-ended or monotonous responses.

3 LUNAR

In this section, we introduce LUNAR method (§3.1) and its layer selection strategy (§3.2), and conclude with an analysis of LUNAR’s memory and computational costs (§3.3). The algorithm pseudo-code can be found in Appendix A.

3.1 Unlearning via Neural Activation Redirection

Previous works [38, 33] have shown that contrastive features can be delineated by computing the ‘steering vector’: $\mathbf{r} = \bar{\mathbf{a}}(x) - \bar{\mathbf{a}}(y)$, i.e., the difference in mean residual stream activations $\bar{\mathbf{a}}$ between

pairs of positive x and negative y examples of contrastive features. These steering vectors have significant implications for influencing model behavior. For instance, a ‘steering vector’ computed out of a contrastive pair of harmful versus harmless prompts can be added to the residual stream activations of harmful prompts to circumvent the model’s safety guardrails [2].

However, given the remarkable ability of transformer architectures to aggregate information and capture abstract representations through high-dimensional residual stream activations, particularly in intermediate layers [13, 9], we conjecture that it is not strictly necessary for two features to be explicitly contrastive in a human-comprehensible sense to compute and utilize ‘steering vectors’. Instead, those can be employed more generally to map a shared hidden feature underlying one group of prompts (i.e., the source feature abstracted by the transformer in intermediate layers) to another group of prompts (i.e., the target feature). We term this process ‘*Activation Redirection*’. This mapping can effectively trigger the model to resemble the behavior associated with the target feature.

In the context of LLM unlearning, the objective is to create an unlearned model that closely mimics the behavior of a retrained model, which explicitly and lively communicates its inability to respond to prompts related to the forget set. To achieve this, we redirect the activations of forget set across all token positions to activations representing the state of inability as follows:

$$\mathbf{a}'^{(l)}_f(x) \leftarrow \mathbf{a}^{(l)}_f(x) + \mathbf{r}^{(l)}_{UV} \quad (4)$$

where $\mathbf{r}^{(l)}_{UV}$, the *unlearning vector (UV)* as a linear intervention in the residual stream activations, is defined as below:

$$\mathbf{r}^{(l)}_{UV} = \frac{1}{|D_{\text{ref}}|} \sum_{x \in D_{\text{ref}}} \mathbf{a}^{(l)}(x) - \frac{1}{|D_f|} \sum_{x \in D_f} \mathbf{a}^{(l)}(x) \quad (5)$$

In Eq. 5, D_f is the forget set and D_{ref} is a set of reference prompts associated with the target feature. Note that D_{ref} can be irrelevant to the unlearning task (i.e., forget set data) and is not restricted to one fixed concept. In one instance, provided the base model is safety-aligned, D_{ref} can be the prompts that activate the model’s internal safety mechanisms to state its inability to positively engage with the unlearned queries. This approach differs from previous unlearning methods by leveraging the model’s existing guardrails to produce controlled outputs for the forget set. Alternatively, we observed that the latest LLMs are well aligned to express knowledge gaps when asked questions about fictitious entities (such as ‘What is the capital of the country \$7&a#!\$’). Here, D_{ref} can be a set of such questions. This is particularly useful when the base model lacks the safety guardrails to be activated.

We then compute the redirected activation based on whether the data are in the forget or remain set, and define the LUNAR loss, $\mathcal{L}_{\text{LUNAR}}$.

$$\mathbf{a}'^{(l)} = \begin{cases} \mathbf{a}^{(l)}(x) + \mathbf{r}^{(l)}_{UV} & \text{if } x \in D_f \\ \mathbf{a}^{(l)} & \text{if } x \in D_r \end{cases} \quad (6)$$

$$\mathcal{L}_{\text{LUNAR}} = \mathbb{E}[\|\mathbf{a} - \mathbf{a}'^{(l)}(x)\|_2] \quad (7)$$

Building on prior work that identified the pivotal role of MLPs in knowledge storage [35], we further propose parameter updates to be **limited to a single down-projection matrix** for effective activation redirection and thus unlearning. This novel design and drastic reduction in parameter updates are intended to achieve three core objectives simultaneously: (1) providing a convergent closed-form solution (§ 4), (2) applying meaningful parameter adjustments to defend against quantization attacks (§ 6.4), and (3) significantly reducing memory and computational costs (§ 3.3).

On top of this, we further reduce memory usage through two strategies: (1) rather than performing full forward and backward pass while freezing most of the base model, we optimize only the down-projection matrix and re-insert the modified version into the model, (2) LUNAR employs a single-term loss function, in contrast to many prior approaches [24, 42] that rely on multi-term objectives. This further minimizes memory consumption during optimization.

3.2 Layer Selection

As part of the LUNAR unlearning process outlined in the subsection above (specifically, after having obtained the unlearning vector and prior to optimization), we identify the optimal intervention layer

by considering two primary objectives: (1) the model should most effectively state its inability to respond, and (2) the response conveys the correct reason.

To assess the first objective, prior work computes a binary refusal score by string-matching common ‘refusal substrings’ (e.g., “I’m sorry” or “As an AI”) [44, 23, 29] or uses the probability of ‘I’ as the first token as a proxy for refusal [2]. However, the substring-matching approach may fail to evaluate the lexical or semantic coherence of the responses [15, 34, 40], while we found the token-probability method can lead to gibberish-like responses of multiple ‘I’s as the probability of ‘I’ increases. Thus, we propose an alternative approach: we compute sentence-level embeddings of the unlearned model’s responses to the forget set and maximize their cosine similarity (s_1) with a list of desired responses⁴. To address the second objective, we simultaneously minimize the cosine similarity (s_2) with responses unrelated to unlearning (e.g., harmfulness, danger, or unethicity). Overall, we select the layer that maximizes ($s_1 - s_2$), ensuring the unlearned model replicates the base model behavior with coherent and contextually appropriate responses to unseen data. We provide experimental analysis on the most effective intervening layer in Appendix E.

3.3 Memory and Computational Costs

The cost of unlearning methods is critical for determining their adoption. Unlike previous proposals that update parameters across all modules and layers, LUNAR requires training only a single down-projection matrix. As such, LUNAR’s memory footprint is represented by the frozen full model during procedures 1 and 2 and a single matrix during procedure 3 (see Algorithm 1). This extreme reduction of the trainable parameters goes beyond a lower impact on the memory, resulting in significant computational efficiency. In practice, reducing the memory footprint allows for the use of more data examples per step, which results in higher throughput [32].

We compare the number of trainable parameters between LUNAR and previous proposals, denoted as N_{LUNAR} and N_{baseline} respectively, with LoRA applied in both cases. $N_{\text{baseline}} = \mathcal{O}(L \cdot m \cdot r \cdot 2d)$, where L is the number of layers, m is number of modules per layer, r is the LoRA rank, d is the dimensionality of each module. Meanwhile, LUNAR requires training only one LoRA module ($m = 1$) on one layer ($L = 1$) such that $N_{\text{LUNAR}} = \mathcal{O}(r \cdot 2d)$.

As in previous works [21], assuming standard optimization conditions, the computational cost per token (FLOPs/token) C for training an LLM is estimated as $C \approx 2N_{\text{fwd}} + 4N_{\text{bwd}}$, where N_{fwd} is the total number of parameters in the forward pass and N_{bwd} is the trainable (non-embedding) parameters in the backward pass. Baselines execute forward and backward pass at a total cost of $C_{\text{baseline}} \approx (2N_{\text{model}} + 4N_{\text{baseline}}) \cdot n_{\text{epoch}}$, where n_{epoch} is the number of training epochs. LUNAR during the first two procedures (see Algorithm 1) executes a forward pass **only once** on the full frozen model at a cost of $C_{\text{LUNAR}|1,2} = 2N_{\text{model}}$. For the third step of LUNAR (see Algorithm 1) (i.e., training down-projection matrix), the FLOPs per token can be estimated as $C_{\text{LUNAR}|3} = 6 \cdot N_{\text{LUNAR}} \cdot n_{\text{epoch}}$. Therefore, the total cost for LUNAR is $C_{\text{LUNAR}} \approx 2N_{\text{model}} + 6 \cdot N_{\text{LUNAR}} \cdot n_{\text{epoch}}$. With the configuration of Llama2-7B as an example (using typical settings of $r = 8$ and $n_{\text{epoch}} = 20$), it is straightforward to show that LUNAR reduces the forward pass cost by $2N_{\text{model}} \cdot (n_{\text{epoch}} - 1)$ and achieves over 100x reduction in backward pass cost, yielding an overall computational cost reduction of approximately $20\times$ compared to baseline methods.

4 Analytical Solution and Convergence Study

In transformer architectures, the down-projection layer functions as a fully connected layer without activation functions. By framing the optimization objective for this layer with $\mathcal{L}_{\text{LUNAR}}$, a convergent closed-form solution can be derived analytically.

Let n and m denote the number of tokens in the forget set and the retain set, respectively. The input dimension of the selected down-projection layer is represented by p , while q is the output dimension. Hidden states before the down-projection layer are therefore $H_f = [h_{1,f}^T, h_{2,f}^T, \dots, h_{n,f}^T] \in \mathbb{R}^{n \times p}$ for the forget set and $H_r = [h_{1,r}^T, h_{2,r}^T, \dots, h_{m,r}^T] \in \mathbb{R}^{m \times p}$ for the retained set, where $h_{i,f}^T$ and $h_{i,r}^T$ are p -dimensional vectors representing each token in the forget and retained set respectively.

⁴This list, carefully curated by observing how base models respond to unseen data (e.g., ‘I apologize that I don’t have access to this information’), will be released upon paper acceptance

Let the original MLP output activations be $A_f^{\text{origin}} = [a_{1,f}^T, a_{2,f}^T, \dots, a_{n,f}^T] \in \mathbb{R}^{n \times q}$ and $A_r^{\text{origin}} = [a_{1,r}^T, a_{2,r}^T, \dots, a_{m,r}^T] \in \mathbb{R}^{m \times q}$. LUNAR introduces a redirection in the activation space for the forget set, resulting in $A_f = [a_{1,f}^T + r_{UV}^T, a_{2,f}^T + r_{UV}^T, \dots, a_{n,f}^T + r_{UV}^T]$, while the activations for the retained set remain unchanged, i.e., $A_r = [a_{1,r}^T, a_{2,r}^T, \dots, a_{m,r}^T]$.

The objective is to optimize the weights of down-projection layer W_{out}^l to minimize the distance between the redirected MLP output and the original output, as follows:

$$\widehat{W} = \arg \min_W ||[H_f, H_r]W - [A_f, A_r]||_2 \quad (8)$$

One can show that there exists a unique solution in the following form: (Proofs of the closed-form solution B.1 and the associated Lemma 1 provided in Appendix B):

$$\widehat{W} = ([H_f, H_r]^\top [H_f, H_r] + \lambda I)^{-1} [H_f, H_r]^\top [A_f, A_r] \quad (9)$$

It is worth noting that the computational cost for Eq. (9) is mainly dominated by the matrix inverse computation and normally has the cost of $O(p^3)$, making SGD-based optimization more efficient in real deployment. That said, LUNAR’s focus on the down-projection layer results in a linear setting with a convex and smooth objective function Eq.(8) (proofs provided in Appendix B.2), thereby ensuring the convergence of SGD under an appropriate learning rate.

5 Experiment Setup

We propose a novel, robust, and efficient method for LLM unlearning. In this section, we conduct experiments to evaluate LUNAR’s performance, focusing on the following research questions:

- RQ1** Does LUNAR improve unlearning efficacy while maintaining model utility? (§6.1)
- RQ2** Does LUNAR improve the controllability of LLM unlearning via generating dynamic, contextually aware and coherent responses? (§6.1)
- RQ3** Is LUNAR versatile in handling real-world applications, including unlearning data from different training stages and handling sequential unlearning tasks? (§6.2 and §6.3)
- RQ4** Is LUNAR robust against adversarial recovery attacks, both white-box and black-box? (§6.4)

5.1 Experimental Setup

Datasets We evaluate LUNAR’s effectiveness on unlearning instance-level knowledge from both fine-tuned models (SFT data) and base models (pre-trained data). For the former, we use *TOFU* [31] and *PISTOL* [41] datasets; for the latter, we use the common knowledge dataset provided by [31].

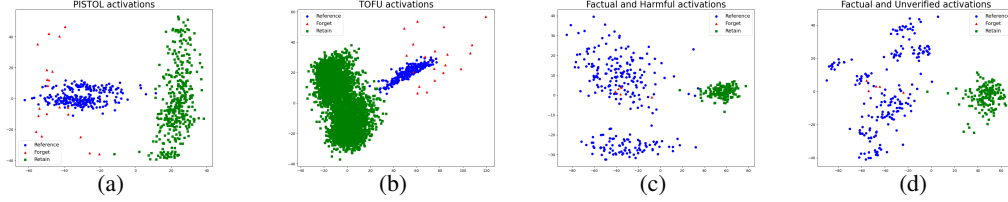
To redirect activations, we use either *harmful prompts dataset* [3] to activate the base model’s internal safety guardrails or *unverifiable prompts dataset*, which we composed using GPT-4 consisting of 200 questions about fictitious objects (e.g., non-existent countries, laws, etc.), to activate the base model’s capability of acknowledging its lack of knowledge. More details can be found in Appendix C.1.

Metrics To evaluate unlearning effectiveness, we define *Deviation Score*: $DS = 100 \times [\text{ROUGE1}_{\text{forget}}^2 + (1 - \text{ROUGE1}_{\text{retain}})^2]^{1/2}$ which takes into account the competing objectives of forget efficacy and retain model utility. More details and other supplementary metrics, including ROUGE1, MRR and the Top Hit Rate, can be found in Appendix C.2.

Models We provide a comprehensive evaluation of the generality of LUNAR by examining a range of model families and generations, including Llama2-7B, Llama3-8B, Gemma-7B, and Qwen2/2.5-7B, encompassing models aligned via Preference Optimization (PO) and Fine-Tuning (FT) [34].

Unlearning Baselines We compare LUNAR against (1) gradient-based methods: Gradient Ascent (GA) [19, 54], Gradient Difference (GD) [26], and GA with KL regularization (UKL); (2) preference optimization (PO)-based methods: Direct Preference Optimization (DPO) [42] and Negative Preference Optimization (NPO) [58]; (3) Representation Misdirection method (RMU) [24] and (4) ‘retrain from scratch’ (a form of exact unlearning), which fine-tunes the base model using only the retain dataset. Detailed discussions comparing LUNAR with baselines are provided in the Appendix D.

Figure 2: PCA visualization of activation space post LUNAR unlearning: (a) unlearn edge AB from the PISTOL dataset; (b) unlearn the first author from the TOFU dataset; (c) unlearn factual dataset from base model with reference dataset be the harmful dataset; (d) unlearn factual dataset from base model with reference dataset be the unverifiable dataset. Base model and PISTOL/TOFU SFT models are Llama2-based.



Optimization We optimize $\mathcal{L}_{\text{LUNAR}}$ (Eq. 6) using all forget data points and an equal number of randomly sampled retain data points, a setting that we find to be sufficient empirically. This also represents a practical choice, as the retain set is typically much larger.

6 Results

6.1 Unlearning Performance

Table 2 shows that LUNAR achieves SOTA unlearning performance, as evidenced by lower deviation scores (up to 11.7x reduction on the PISTOL dataset with Gemma-7B model) and superior control scores. Examples in Table 1 and Appendix F.1 further visualize LUNAR’s superior controllability, significantly reducing hallucinations and improving the coherent expression of its inability to respond within the conversational context. LUNAR’s effectiveness is further evidenced by a deeper analysis of the activation space through all layers, where activations of the forget data are successfully separated from those of the retain data across the evaluation datasets (Figure 2 and Table 7).

Interestingly, we also found that **fine-tuning with the retained set (a form of exact unlearning) does not guarantee sufficient content regulation**, as unlearned knowledge can be reintroduced in-context, allowing the model to behave as if it retains the forgotten knowledge. This echoes with arguments in [47]. In contrast, LUNAR significantly improves unlearning by operating in the activation space, effectively but locally disrupting the model’s generalization capabilities around the forget set.

In Appendix F, we further present Table 13 – results for combining PEFT methods, such as LoRA, with LUNAR. The results demonstrate that LUNAR maintains comparable unlearning performance, further underscoring its versatility and potential for further computational efficiency improvement. Additionally, Table 12 shows that the LUNAR unlearned model maintains the base model’s performance on downstream tasks, confirming that LUNAR performs targeted, minimally invasive interventions that removes specific knowledge without degrading general model capabilities.

6.2 Unlearning Pre-trained Data from Base Models

We observe that modern LLMs exhibit an ability to express a lack of knowledge when presented with fictitious or unverifiable questions. This ability is often stronger in pre-trained models compared to SFT models [46]. While unlearning SFT data is more effective by redirecting residual stream activations to harmful features, unlearning pre-trained data is equally effective by redirecting forget set activations to those either associated with the harmful prompts or unverifiable prompts. The effectiveness of LUNAR in unlearning pre-trained data is presented in Table 3.

6.3 Unlearning Sequentially

Another practical scenario in LLM unlearning deployment involves private data being removed incrementally over time, as unlearning requests arrive sequentially. Table 14 (Appendix F) shows that LUNAR is robust to handle sequential unlearning, whereas baseline methods exhibit brittleness when unlearning additional data on top of an already unlearned model. LUNAR consistently achieves strong results across different models, comparable to the performance observed in single-round unlearning.

Table 2: Comparison of LUNAR’s unlearning performance with retraining and baseline methods across models. Metrics are marked with $\uparrow$ (higher is better) and $\downarrow$ (lower is better); best results in **bold**. Note that PISTOL provides a clearer evaluation due to concise ground truth, while TOFU’s open-ended QAs lead LUNAR to generate contextual tokens, increasing ROUGE1-based deviation scores despite effective expression of its lack of knowledge. Results on newer model generations in the Appendix confirm LUNAR’s generalizability.

Method	Llama2-7B			Gemma-7B			Qwen2-7B		
	Deviation Score (DS) $\downarrow$	Compare to Best DS	Control Score $\uparrow$	Deviation Score (DS) $\downarrow$	Compare to Best DS	Control Score $\uparrow$	Deviation Score (DS) $\downarrow$	Compare to Best DS	Control Score $\uparrow$
PISTOL									
Retrain	34.1	4.4x	0.355	26.1	4.1x	0.358	33.0	5.5x	0.356
GA	52.4	6.7x	0.353	57.6	9.1x	0.351	32.7	5.5x	0.359
GD	54.9	7.0x	0.355	35.5	5.6x	0.358	30.6	5.2x	0.358
UKL	54.3	7.0x	0.394	73.5	11.7x	0.352	54.4	9.1x	0.348
DPO	22.8	2.9x	0.524	23.4	3.7x	0.692	24.6	4.1x	0.594
NPO	39.8	5.1x	0.352	26.6	4.2x	0.359	30.7	5.2x	0.353
RMU	58.6	7.5x	0.351	38.3	6.1x	0.341	60.4	10.2x	0.348
LUNAR	7.8	1.0x	0.677	6.3	1.0x	0.701	5.9	1.0x	0.640
TOFU									
Retrain	31.7	2.1x	0.429	32.5	2.4x	0.425	36.1	2.4x	0.402
GA	40.7	2.7x	0.456	49.6	3.7x	0.460	27.5	1.9x	0.383
GD	37.2	2.5x	0.453	49.6	3.7x	0.462	25.2	1.7x	0.422
UKL	60.6	4.1x	0.361	86.0	6.4x	0.402	74.5	5.0x	0.401
DPO	15.2	1.0x	0.515	20.2	1.5x	0.588	60.7	4.1x	0.433
NPO	33.4	2.2x	0.509	44.4	3.3x	0.487	26.7	1.8x	0.477
RMU	64.8	4.3x	0.429	62.1	4.7x	0.399	69.1	4.8x	0.406
LUNAR	14.9	1.0x	0.608	13.1	1.0x	0.659	14.3	1.0x	0.609

Table 3: Performance of unlearning individual factual data points from base models demonstrates that activation redirection is effective using either harmful or unverifiable prompts as D_{ref} in Eq. 5.

Model	D_{ref}	Forget ROUGE1 $\downarrow$	Retain ROUGE1 $\uparrow$	Control Score $\uparrow$
Llama2-7B	Harmful	0.000	0.981	0.694
	Unverifiable	0.000	0.986	0.654
Llama3-8B	Harmful	0.000	0.981	0.673
	Unverifiable	0.000	0.976	0.620
Gemma-7B	Harmful	0.000	0.859	0.671
	Unverifiable	0.000	0.859	0.714
Qwen2-7B	Harmful	0.000	0.977	0.683
	Unverifiable	0.000	0.980	0.625
Qwen2.5-7B	Harmful	0.000	0.996	0.653
	Unverifiable	0.000	0.987	0.675

Table 4: Attack performance comparing different models and attack methods on the PISTOL dataset (ROUGE1 of the forget set). The Layer Skip and Reverse Direction attacks bypass or reverse activation redirection layers, respectively. Quantization applies 4-bit precision to the full model.

Model	LUNAR (Top-K)	Layer Skip	Reverse Direction	4-bit Quant.	Prompt Paraphrase
Llama2-7B	0.007	0.117	0.000	0.167	0.019
Llama3-8B	0.070	0.180	0.000	0.123	0.031
Gemma-7B	0.060	0.150	0.000	0.060	0.036
Qwen2-7B	0.012	0.115	0.160	0.000	0.025
Qwen2.5-7B	0.183	0.100	0.000	0.000	0.032

6.4 Robustness Study

In this section, we show LUNAR is robust to white and black-box attacks, where the former operates under strong assumptions that the attacker *at least* possesses full knowledge of the model weights.

Layer skip attack For a white-box deployed model, the layer skip attack is designed to bypass the intervention layer(s), which can be effective given the ensemble nature of transformer architectures [51, 7]. In this scenario, performing activation redirection on multiple layers (identify top- K layers through the layer selection process) serves as an effective defense. For Llama2-7B, selecting top- K ($K = 3$) layers is an effective defense with ROUGE1 score only increasing marginally to about 0.1 (Table 4), indicating minimal recovery of unlearned information. A closer examination of generated outputs reveals this minor increase primarily stems from two factors: (1) unigram matches between generated text and ground truth rather than accurate responses in their entirety, and (2) questions with binary choices where the model occasionally guesses correctly (refer to examples of post-attack responses in Appendix F.3). Overall, the unlearned model remains non-usable on the forget set, underscoring the robustness of LUNAR against such attacks.

Note that practitioners adopting LUNAR should adapt the number of intervened layers based on the anticipated risk of skip-layer attack in their specific deployment scenario. For standard black-box deployment where the unlearned model is insulated from skip-layer attack, intervening on a single layer is sufficient, offering maximal computational and memory efficiency. Conversely, in scenarios